Advanced Programming in the UNIX Environment

Week 10: HTTP

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HTTP

The Hypertext Transfer Protocol — A simple request/response protocol.

- RFC1945 (HTTP/1.0; May 1996)
- RFC2616 (HTTP/1.1; June 1999)
- RFC7540 (HTTP/2; May 2015)
- RFC9114 (HTTP/3; June 2022)

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The Hypertext Transfer Protocol

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1. client sends a request to the server
   - **request method**

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GET
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The Hypertext Transfer Protocol

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   - **URI**

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   - URI
   - **protocol version**

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2. server responds
   - **status line (including success or error code)**

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GET / HTTP/1.0
If-Modified-Since: Mon, 13 Nov 2023 18:03:39 GMT
User-Agent: Mozilla/5.0

HTTP/1.0 200 OK
Date: Mon, 13 Nov 2023 19:21:10 GMT
Server: gws
Content-Type: text/html; charset=ISO-8859-1

<!doctype html><html itemscope=""
itemtype="http://schema.org/WebPage"
lang="en"><head><meta
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HTTP is a request/response protocol:

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   - status line (including success or error code)
   - **server information**
   - entity meta-information
   - content

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The Hypertext Transfer Protocol

Server status codes:

- `1xx` – Informational; Request received, continuing process
- `2xx` – Success; The action was successfully received, understood, and accepted
- `3xx` – Redirection; Further action must be taken in order to complete the request
- `4xx` – Client Error; The request contains bad syntax or cannot be fulfilled
- `5xx` – Server Error; The server failed to fulfill an apparently valid request

The Hypertext Transfer Protocol

```
$ telnet www.stevens.edu 80
[...]
GET / HTTP/1.1
Host: www.stevens.edu

HTTP/1.1 301 Moved Permanently
Date: Mon, 13 Nov 2023 20:10:40 GMT
Transfer-Encoding: chunked
Connection: keep-alive
Cache-Control: max-age=3600
Expires: Mon, 13 Nov 2023 21:10:40 GMT
Location: https://www.stevens.edu/
Server: cloudflare
```

The Hypertext Transfer Protocol

```
$ openssl s_client -crlf -servername www.stevens.edu -connect www.stevens.edu:443
[...]
GET / HTTP/1.1
Host: www.stevens.edu

HTTP/1.1 200 OK
Date: Mon, 13 Nov 2023 20:12:48 GMT
Content-Type: text/html; charset=utf-8
Transfer-Encoding: chunked
Connection: keep-alive
Cache-Control: public, max-age=0, must-revalidate
Strict-Transport-Security: max-age=15552000
```

The Hypertext Transfer Protocol

HTTP is a *Transfer* Protocol – serving data, not any specific text format.

- `Accept` or `Accept-Encoding` client header can specify different formats (e.g., `application/json`) or encodings such as *gzip*, Brotli etc.
- corresponding server headers: `Content-Type` and `Content-Encoding`

Data can be transferred in either direction.

- POST, PUT, DELETE
- GET, HEAD
- CONNECT, OPTIONS

HTTP - more than just static data

HTTP is a Transfer Protocol – what is transferred need not be static; resources may generate different data to return based on many variables.

- CGI – resource is executed, needs to generate appropriate response headers
- server-side scripting (ASP, PHP, Perl, …)
- client-side scripting (JavaScript/ECMAScript/JScript,…)
- applications based on HTTP, using:
  - AJAX
  - RESTful services
  - JSON, XML, YAML to represent state and abstract information

HTTP in your final project

- protocol versions supported: `1.0`
- request methods supported: `GET`, `HEAD`
- request headers supported: `If-Modified-Since`
- response headers included: `Date`, `Server`, `Last-Modified`, `Content-Type`, `Content-Length`

HTTP in your final project

Let’s create your building blocks:

- program startup and dæmonization
- rough network logic
- input / request parsing and validation
- cgi handling
- user directory handling
- directory index generation
- regular file serving

Links

- https://developer.mozilla.org/en-US/docs/Web/HTTP
- https://www.w3.org/Protocols/
- https://www.rfc-editor.org/rfc/rfc1945.html
- https://stevens.netmeister.org/631/f23-group-project.html
- https://www.netmeister.org/cgi-bin/env.cgi